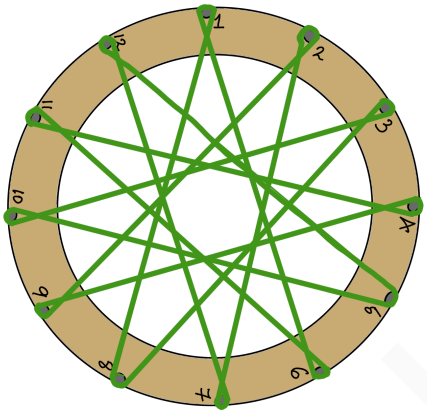
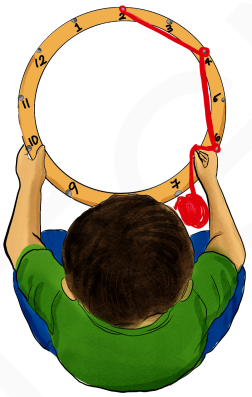
String Art

Grade 6: Prime Time

Introduction

Imagine using numbers and their relationships to create beautiful artworks. In this activity, we will take a circle with 12 equally spaced points. By connecting these points using a fixed rule, we will create beautiful shapes such as regular polygons and stars.





This activity helps students understand:

- Factors and multiples of a number

Activity	Time	Description
Launch	5 min	Teacher demonstrate the activity Supporting Links: Activity Video: https://tinyurl.com/stringartactivityvideo
Trail by Students	15 min	Students Tryout the activity with each other. Student Worksheet: https://tinyurl.com/stringartworksheet
Discussions and Explorations	15 min	Attempting the worksheets and the discussion based on the activity

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CT Components

Algorithmic Thinking:

In this activity, students will follow a rule - Starting from a number on a circle with **12** equally spaced points, add a step number (say **n**) to it and keep doing it until they reach back to the starting point. This becomes the algorithm for every circle.

Pattern Recognition:

Once the different shapes for the circle with different step numbers have been formed, students will try to recognise the shapes formed. They will then try to see a pattern between the shape formed and the step numbers and points on the circle.

Generalisation:

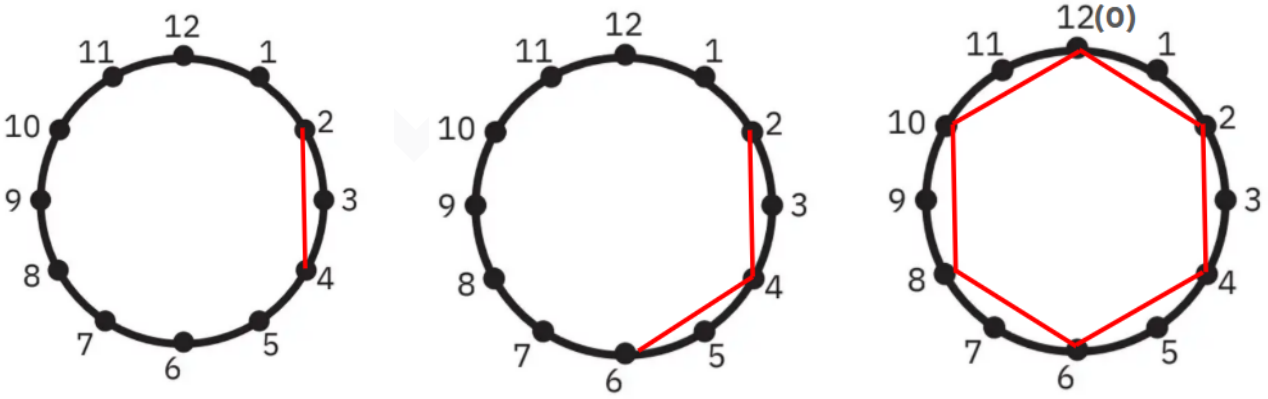
After recognising the pattern of shapes formed on a circle with different step numbers, students will generalise the shapes formed for any circle with **m** equidistant points and **n** steps.

Activity

You are given a circle with 12 equidistant points.
You have to connect these points using a rule.

Rule:

1. Starting from 2, add 2 to it and connect it with the next number, i.e. 4.
2. Add 2 to 4, then connect it to the next.
3. Keep adding 2 until you reach the starting point.



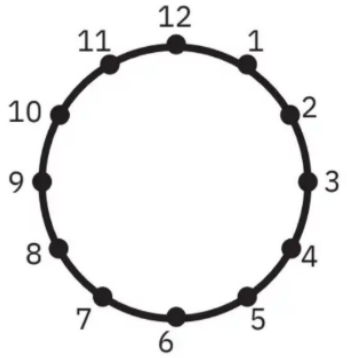
1. How many steps did it take to reach the starting point? What shape do you get?

Answer: We can see that the 6 steps formed equal chords on the circle, thus giving us a regular hexagon. Also, if we add 2 to the last point again, we will just retrace the steps we took in the first sequence.

Competencies: Algorithmic Thinking

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Now starting from 3, add 3 to it and join to the next point. Keep adding 3 until you reach the starting point.



1.3 How many steps would be needed to reach the starting point?

a) Two
b) Three
c) Four
d) Five

Competencies: Algorithmic Thinking

Answer: Starting from 3 and adding 3 at every step, we proceed like this
3 → 6 → 9 → 12 → 3
After 4 steps, we return to 3.
Shape: Since the 4 steps formed equal chords on the circle, we get a square.

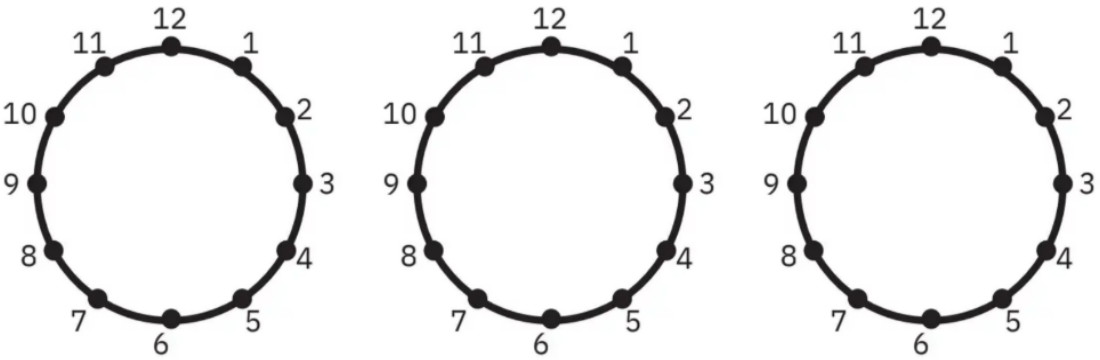
Note: If we add 3 to 3 again, we will just retrace the steps that we made in the first sequence.

<u>Repeat the same process by</u>

a) Starting from 4 and adding 4
b) Starting from 5 and adding 5
c) Starting from 6 and adding 6

Until you reach the starting point.

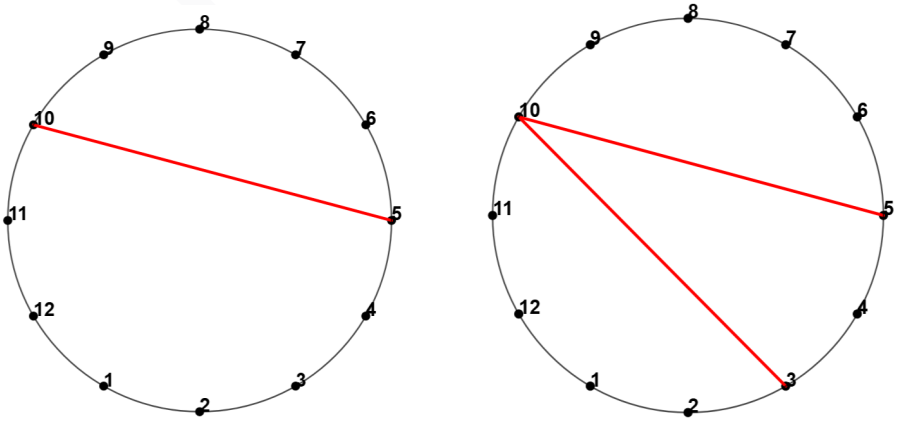
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What shapes do you get? Record your answers in the table below.

Number added	No. of steps till the starting point	Shape formed
2	6	Hexagon
3	4	Square
4		
5		
6		

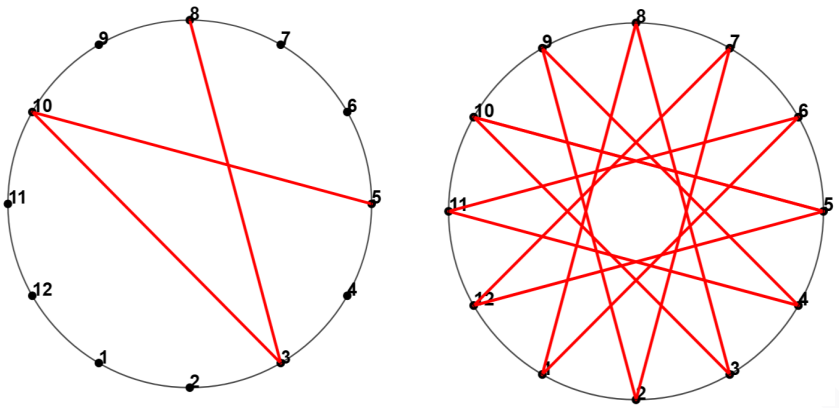
Note: When the starting point is 5, adding 5 to it gives 10. Now, adding 5 to 10 gives 15, which is not on the circle. But since the numbers lie on a circle, we can keep going around it. So 15 will go to 3 (12 + 3 = 15).



Now add 5 to 3, and we get 8. Keep going in the same manner, and we will get this sequence of steps:

5→ 10→3→8→1→6→11→4→9→2→7→12→5

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So after 12 steps, we will reach 5 again.

Do you notice any **patterns** in the **numbers we are adding, the number of points on the circle, and the shape being formed in the circle**?

Why did you get polygons on adding 2, 3 and 4, while a star on adding 5? Is there any relationship between these numbers and 12, the number of points on the circle?

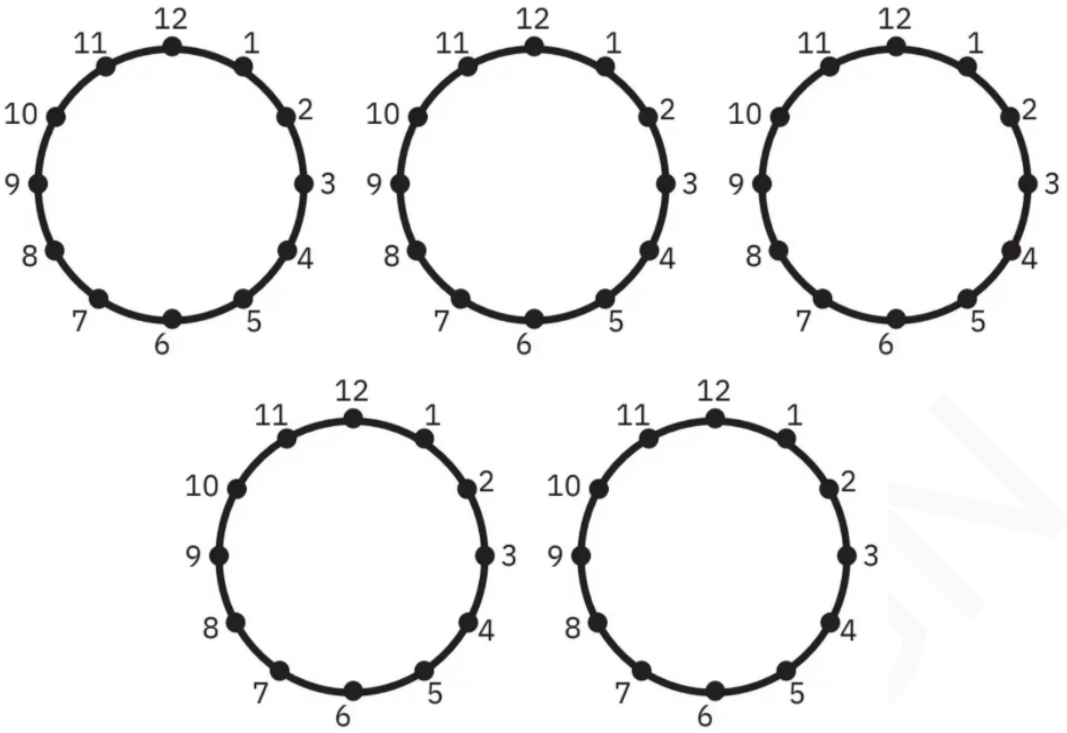
Competencies: Algorithmic Thinking, Pattern Recognition

Explanation: The numbers 2, 3, and 4 are factors of 12, while 5 is not. To reach 12 - 2, 3, and 4 take 6,4 and 3 steps respectively and reach the starting point. But when we start from 5, we have to keep on going until we reach a number that is a common multiple of both 5 and 12, so that the starting point is the same as the ending point. So 5 takes 12 steps, adding to 60, which is a common multiple of both 5 and 12 and in the process, it covers all points, giving us a 12-pointed star.

Explorations

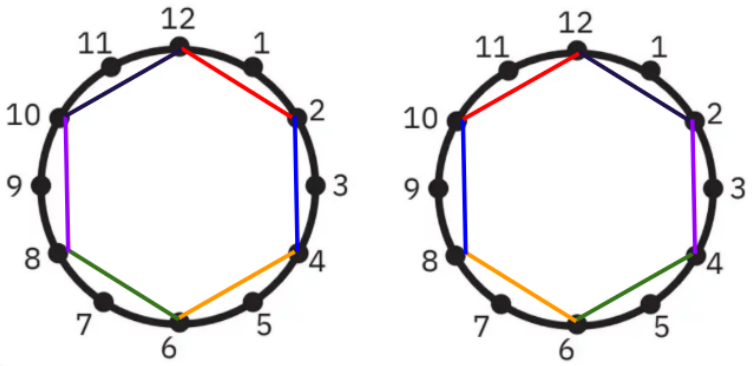
1. What shapes will you get when you add 7, 8, 9, 10 and 11 until you reach the starting point?
   Try to find out the relationship between the numbers and the number of points on a circle.

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Competencies: Algorithmic Thinking, Pattern Recognition, Generalisation

Explanation - We will observe that 7, 8, 9, 10, and 11 will have the same shape as in the cases of 5, 4, 3, 2, and 1, respectively. And if you look at these numbers on the circle, they are the reflections of each other. In case 2, if I mark my lines in order in different colours and then repeat the procedure for 10, we will be able to see that the hexagons obtained are reflections of each other.



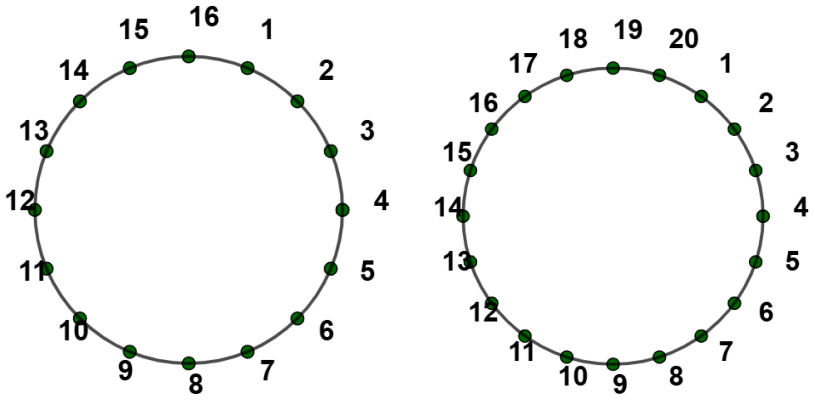
In the first image, we start from 2, the blue string connects to 4. Then 4 connects to 6 through a yellow string, and so on.
In the second image, we start from 10, add 10 to get 20, which will go to 8 (12 + 8), so the first string is blue from 10 → 8. Then 8 plus 10 is 18, which goes to 6 using a yellow string and so on.

So, the numbers that are reflections will have the same shapes.

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We can also generalise - If there are m points on a circle, and we know the shapes formed by step number n, then m-n will also have the same shape.

2. Two circles, one with 16 points and the other with 20 points, are given. Which of the two will give a square when added to which number?



a) The 16-point circle, on adding 4
b) The 16-point circle, on adding 12
c) The 20-point circle, on adding 5
d) The 20-point circle, on adding 15

Competencies: Pattern Recognition, Decomposition

Explanation: All 4 options are correct.
For a 16-point circle, to get a square, we need to take 4 steps. So, 16 divided by 4 is 4, hence adding 4 gives us a square. Also, 16-4 = 12 also gives a square.
We can similarly find a 20-point circle.

For further exploration on the teachers' part: Try to explore the same exercise with a 20-point circle and different step numbers. Try to see what happens when step numbers are not proper factors, but have a common factor with the number of points on the circle.

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